

Assignment 1

due: 2025-09-16

Problem 1: Let F be a field, let I be a set, and let $V = V_I$ the set of finitely supported F -valued functions on I .

- a. Explain why V is an F -vector space.
- b. For $i \in I$, let δ_i be the function $\delta_i : I \rightarrow F$ given by the rule

$$\delta_i(j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{else} \end{cases}$$

Prove that $\mathcal{B} = \{\delta_i \mid i \in I\}$ is a basis for V_I as F -vector space. You need to prove that $\mathcal{B}$ spans V and that $\mathcal{B}$ is linearly independent.

- c. If $f : I \rightarrow J$ is a function, explain why there is a linear mapping $\Phi_f : V_I \rightarrow V_J$ for which

$$\Phi_f(\delta_i) = \delta_{f(i)}$$

for each $i \in I$. What can you say about Φ_f if f is injective? surjective? bijective?

Problem 2: Let F be a field and let V and W be F -vector space.

- a. Suppose that $\varphi : V \rightarrow W$ is an invertible F -linear map. Using φ , define a group isomorphism $G_\varphi : \text{GL}(V) \rightarrow \text{GL}(W)$.
- b. Suppose that $V = W$, so that $\varphi : V \rightarrow V$ is actually an element $\varphi \in \text{GL}(V)$. Show that G_φ is the inner automorphism given by conjugation with φ .